

Business scope

Area	Location-level analysis and filtering
Sales	YOY growth, YTD sales, rolling 12-month sales and units sold
Pricing	Offer vs purchase price, median price change and price per SQM
Property	House type, SQM, SQM price and property age
Market context	Inflation, interest and yield by house type

2. BUSINESS PROBLEM

#	Business problem	Why it matters
1	Sales performance visibility	Management needs a quick way to see whether sales are growing or declining and which sales types drive the change.
2	Regional performance	Different regions have different sales volumes and price movements; weak regions need investigation while strong regions may offer expansion opportunities.
3	Channel performance	Auction, regular, other and family sales may behave differently, but raw data does not make those differences immediately visible.
4	Pricing decisions	Offer price, purchase price and price-per-SQM relationships need to be analyzed to understand pricing behavior.
5	Property segmentation	House types differ substantially in average transaction value, size and market indicators.
6	Data quality & reporting	Raw transaction data contains missing values and requires cleaning, transformation and repeatable reporting.

Business objective: Create a single interactive reporting layer that allows decision-makers to identify growth drivers, detect declining segments, compare regional performance and support pricing and sales strategy.

3. ANALYTICS APPROACH & SOLUTION ARCHITECTURE

Stage	Implementation
1. Ingest	Loaded the housing CSV dataset into Google BigQuery.
2. SQL analysis	Created a test table, inspected records, calculated average purchase price by sales type and performed data updates/validation.
3. Power BI connection	Connected BigQuery to Power BI Desktop for reporting.
4. Data cleaning	Used Power Query to remove relevant null/missing values and prepare fields for analysis.
5. DAX modeling	Created sales growth, rolling sales, YTD, units sold, pricing and segmentation measures/columns.
6. Visualization	Built three dashboard pages: House Market Overview, Sales Performance and House Type Analysis.
7. Deployment	Published the report to Power BI Service with a workspace and scheduled refresh.

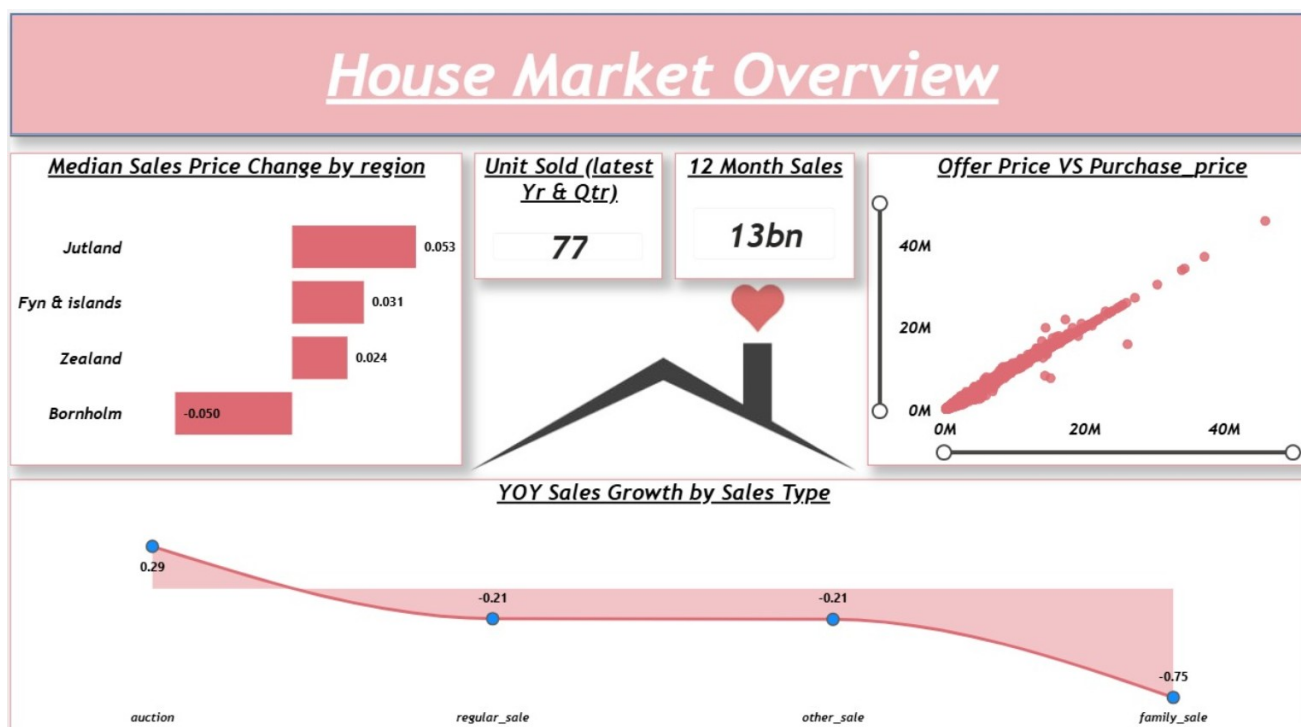
Analytics flow: CSV → BigQuery → SQL → Power BI → Power Query → DAX → Interactive dashboards → Power BI Service → Scheduled refresh.

Key analytical principle: The dashboard is designed to support descriptive and diagnostic analytics. It shows what happened and where differences exist; it does not by itself prove that one factor caused a sales change.

4. KPI FRAMEWORK

KPI	Business purpose
YOY Sales Growth	Measures year-over-year change in purchase sales and compares sales types.
Rolling 12-Month Sales	Tracks sales value over the latest 12-month period; dashboard value $\approx$ 13bn.
Units Sold – Latest Year & Quarter	Distinct property count for the latest year/quarter; dashboard value = 77.
Total YTD Sales	Tracks cumulative sales from the start of the year.
Median Sales Price Change	Compares current-year and prior-year median purchase price by region.
Average Price SQM	Compares property price per square meter across regions.
Offer Price	Derived from purchase price and the dataset's offer-to-purchase percentage change.
Offer-to-SQM Ratio	Compares total offer value with total property area by sales type.
Property Age	Difference between transaction year and construction year.

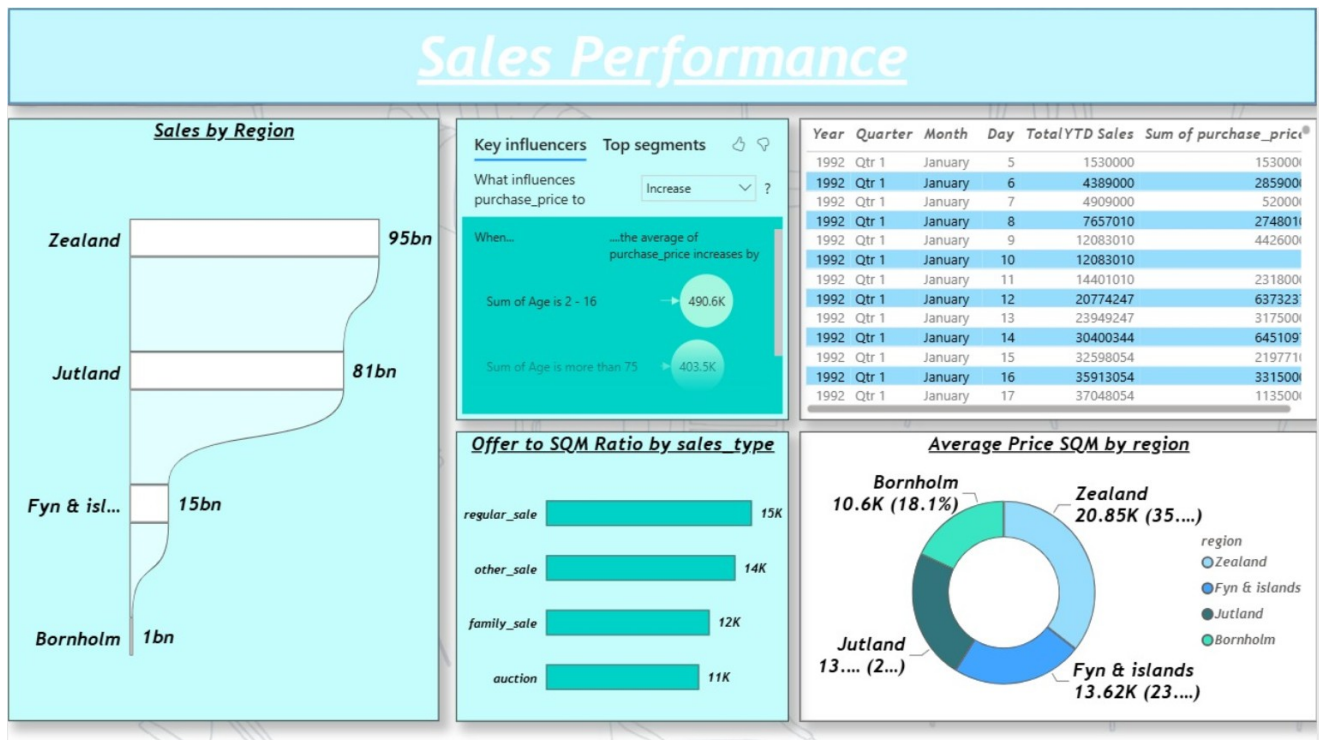
5. DASHBOARD 1 — HOUSE MARKET OVERVIEW



Major finding: Auction sales show +29% YoY growth, while regular sales and other sales each decline by 21%, and family sales decline by 75%. This indicates a highly uneven sales-channel performance.

Segment	Change	Interpretation
Auction	+29%	Only sales type with positive YoY growth; investigate and scale successful characteristics.
Regular sale	-21%	Declining channel; review pricing, inventory and customer demand.
Other sale	-21%	Declining channel; identify property and regional drivers.
Family sale	-75%	Largest decline; requires immediate diagnostic investigation.
Jutland	+5.3%	Highest median sales price change among the four regions.
Bornholm	-5.0%	Negative median price change; review local demand and inventory.

6. DASHBOARD 2 — SALES PERFORMANCE

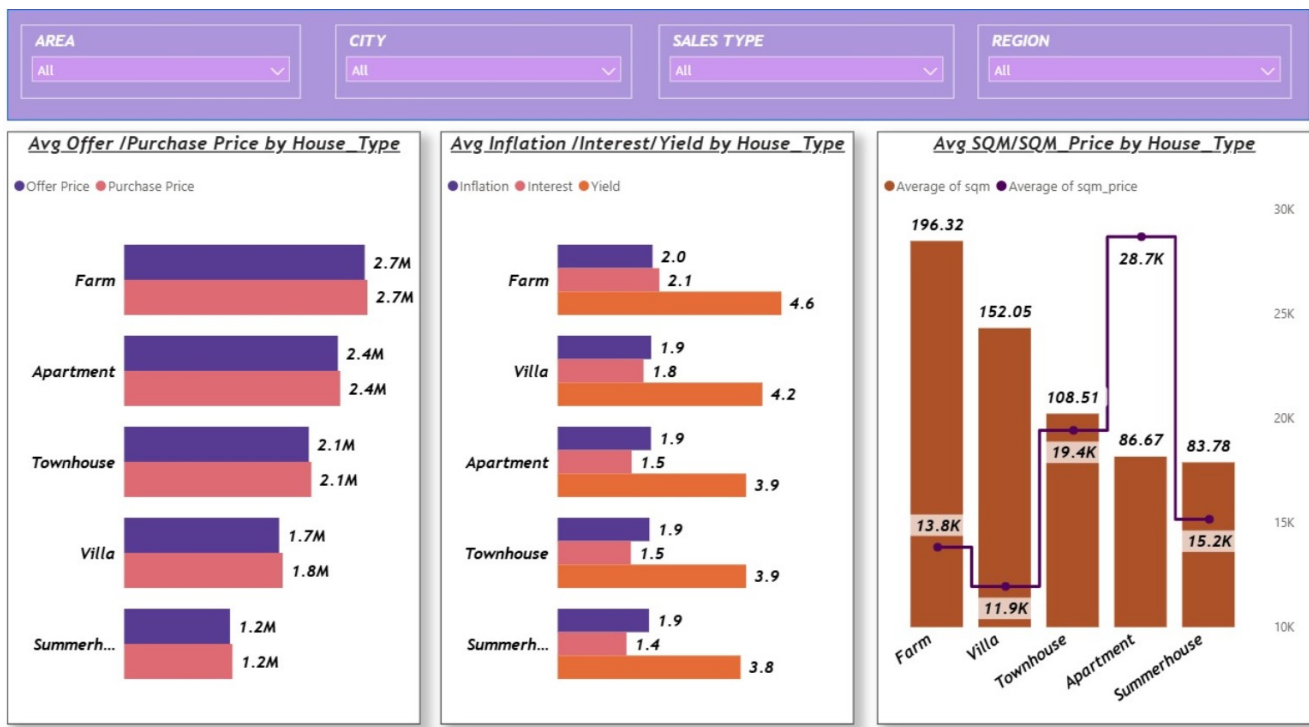


Regional sales volume: Zealand contributes approximately 95bn in sales, followed by Jutland at 81bn, Fyn & Islands at 15bn, and Bornholm at 1bn. This shows a strong concentration of sales in Zealand and Jutland.

Recent activity: The dashboard reports 77 units for the latest year/quarter KPI and approximately 13bn in rolling 12-month sales. These KPIs provide management with an executive snapshot of transaction activity and recent sales value.

Pricing by region: Zealand has the highest average SQM price in the dashboard at approximately 20.85K, followed by Fyn & Islands at 13.62K. This supports differentiated pricing and market strategy by region.

7. DASHBOARD 3 — HOUSE TYPE ANALYSIS



Property-value segmentation: Farms have the highest average purchase price at approximately 2.7M, followed by apartments at 2.4M, townhouses at 2.1M, villas at 1.8M, and summerhouses at 1.2M.

Sales-channel pricing: Offer-to-SQM ratio is highest for regular sales at about 15K, followed by other sales at 14K, family sales at 12K, and auction at 11K. This provides a useful benchmark for pricing by sales type.

Economic context: Inflation, interest and yield were compared across house types. These indicators should be used as market context when explaining price and sales movements, rather than as proof of causation.

8. KEY OUTCOMES OF THE ANALYSIS

Outcome	Quantified result	Business meaning
Sales-channel growth	+29% auction YoY	Auction is the only channel in the dashboard with positive YoY growth.
Largest decline	-75% family sale YoY	Family sales require the highest-priority diagnostic review.
Other channel declines	-21% regular; -21% other	Traditional/other channels are weakening and need corrective action.
Regional price growth	+5.3% Jutland	Jutland shows the strongest median sales-price change.
Regional price decline	-5.0% Bornholm	Bornholm needs local market/inventory investigation.
Sales concentration	95bn Zealand; 81bn Jutland	Two regions account for the largest reported sales values.
Rolling sales	≈13bn	Recent 12-month sales value provides a current performance benchmark.
Latest activity	77 units	Latest year/quarter transaction KPI.

Important interpretation: The available project metrics are sales/transaction and pricing metrics. A true profit percentage cannot be calculated from the supplied dashboard because the dataset/report does not provide a reliable cost, expense, acquisition-cost or margin field. Therefore, the percentages above represent sales growth/decline and price movement, not company profit or net profit.

9. PROPOSED BUSINESS SOLUTION

1. Scale the auction growth engine

Auction sales grew 29% YoY. Identify the regions, cities, house types and price bands behind this growth and test whether the same characteristics can be expanded.

2. Recover declining channels

Regular and other sales fell 21%, while family sales fell 75%. Build a root-cause review around pricing, inventory mix, geography, property age and customer demand.

3. Protect high-value markets

Zealand has the highest reported sales volume at about 95bn, while Jutland shows the strongest median price growth at 5.3%. Use separate volume and growth strategies for these markets.

4. Address weak regions

Bornholm shows a 5.0% median price decline and only about 1bn in sales in the dashboard. Review local demand, property mix and pricing before increasing inventory.

5. Introduce data-driven pricing

Use offer price, purchase price, SQM and SQM price together to establish pricing benchmarks by region, house type and sales channel.

6. Segment inventory

Prioritize premium property strategies for higher-value types such as farms and apartments, while using volume/affordability strategies for lower-ticket segments such as summerhouses.

7. Operationalize the dashboard

Keep the Power BI Service report on scheduled refresh and establish a recurring management review of YOY growth, rolling sales, regional price change and channel performance.

Recommended management cadence: weekly review of channel and regional anomalies, monthly review of pricing and inventory mix, and quarterly review of growth strategy and regional expansion.

10. EXECUTIVE CONCLUSION

The housing sales analytics project converts raw transaction data into a decision-support system for sales and pricing management. The analysis reveals that performance is highly uneven: auction sales grew 29% YoY, while family sales experienced a 75% decline. Jutland recorded the strongest median sales-price growth at 5.3%, whereas Bornholm declined by 5.0%. At the sales-volume level, Zealand reported approximately 95bn and Jutland 81bn in the dashboard.

The recommended strategy is therefore not simply to 'increase sales' across the entire business. Instead, management should scale the proven auction growth channel, recover declining sales types, protect high-volume markets, investigate weak regions, and use property-level pricing benchmarks to improve conversion and revenue quality.

The project also establishes a reusable BI architecture—BigQuery + SQL + Power BI + DAX + Power BI Service—that can be extended with profit/margin data, customer segmentation, inventory availability, conversion rates and forecasting in a future phase.

+29%	-75%	+5.3%	-5.0%
Auction sales growth	Family sales decline	Jutland median price change	Bornholm median price change

Final business message: The dashboard identifies where the business is growing, where it is losing momentum, and which segments should receive management attention. The next step is to connect these insights to costs, margins, inventory and conversion data so that sales growth can be translated into measurable profitability.